

SnowMap: A Tool to Support the Snowballing Process in Literature Reviews

Jailma Januário
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
jailma.silva@inf.puc-rio.br

Alice Fico
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
alice2fico@gmail.com

Julia Guimarães Simão
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
emailpucdajulia@gmail.com

Laíza Ágatha
Rodrigues Gomes
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
laizadocs.gomes@gmail.com

Fernanda Piquet
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
nandapiquet@gmail.com

Ricardo Ramalho
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
rjunior@inf.puc-rio.br

Juliana Alves Pereira
Department of Informatics,
PUC-Rio
Rio de Janeiro, Brazil
juliana@inf.puc-rio.br

Abstract

Context: Snowballing is a widely adopted strategy in Systematic Literature Reviews for identifying relevant studies through citation relationships. However, manual execution of forward and backward snowballing is time-consuming, difficult to scale, and highly dependent on fragmented bibliographic data sources. **Goal:** This study presents SnowMap, a tool designed to support automated forward and backward snowballing, as well as downstream SLR activities such as metadata analysis and LLM-based study screening. **Method:** SnowMap integrates multiple bibliographic APIs and citation retrieval mechanisms to automatically identify related studies from a set of seed papers. In addition, the tool supports metadata processing and automated screening using Large Language Models. We evaluated the tool in realistic SLR scenarios by analyzing metadata completeness, publication venue characteristics, and the effectiveness of different prompting strategies for LLM-based screening. **Results:** The results indicate that SnowMap is capable of supporting automated and reproducible snowballing workflows, achieving high consistency for structural metadata such as DOI and publication year. However, descriptive metadata showed greater variability, particularly in backward snowballing. Furthermore, chain-of-thought prompting achieved better performance in study screening tasks when compared to other prompting strategies. **Conclusion:** SnowMap contributes to the automation of snowballing-based review workflows by integrating citation retrieval, metadata analysis, and LLM-assisted screening into a unified framework, while also highlighting challenges related to metadata quality, scalability, and dependency on external bibliographic sources.

Demo video:<https://youtu.be/mZ2NZ-xrWLo>

Keywords

Systematic Literature Reviews, Snowballing, LLMs, Automation

1 Introduction

Snowballing is a widely adopted strategy in Systematic Literature Reviews (SLRs) for identifying relevant studies through citation relationships. This strategy is commonly divided into two directions: backward snowballing, which analyzes the reference lists of selected

studies to identify prior work, and forward snowballing, which identifies studies that cite the selected papers [17].

Despite its usefulness, the manual execution of both forward and backward snowballing is time-consuming, difficult to scale, and prone to inconsistencies, especially when multiple seed studies are involved [17]. These challenges are further exacerbated by the fragmentation of bibliographic data sources and differences in how citation and reference information are represented across platforms [5]. As a consequence, researchers often need to manually combine information from multiple systems, increasing the effort required to maintain traceability and reproducibility throughout the review process.

A previous study [8], which conducted a comparative analysis of tools supporting snowballing in SLRs, showed that existing solutions still provide limited support for automating the review process. The main limitations identified include issues related to metadata completeness and consistency, lack of integration between forward and backward snowballing, low reproducibility of retrieval workflows, and insufficient support for downstream SLR activities, such as study screening and metadata analysis. Furthermore, most existing tools still rely heavily on manual interaction, limiting their applicability in large-scale systematic reviews.

In this context, this paper presents SnowMap, a tool designed to support automated forward and backward snowballing in SLR. The tool integrates multiple bibliographic data sources and citation retrieval mechanisms to automatically identify related studies from an initial set of seed papers. In addition, the tool supports downstream SLR activities, including metadata analysis and study screening based on inclusion and exclusion criteria using Large Language Models (LLMs). By integrating citation retrieval, metadata processing, and screening support into a unified workflow, we aim to improve the efficiency, traceability, and reproducibility of snowballing-based review processes.

The objective of this study is to describe and evaluate SnowMap in realistic SLR scenarios. In particular, we investigate the completeness and consistency of the retrieved metadata, the characteristics of publication venues identified through snowballing, and the effectiveness of different prompting strategies for LLM-based screening tasks. The results provide evidence regarding both the capabilities

and limitations of SnowMap in supporting automated snowballing workflows. In particular, we observe that metadata quality and completeness vary depending on the data source and the type of citation relationship explored, directly affecting the traceability of retrieved studies.

The main contributions of this paper are summarized as follows:

- The design and implementation of SnowMap, a tool for automated forward and backward snowballing in SLRs;
- The integration of multiple bibliographic data sources and citation retrieval mechanisms into a unified and reproducible workflow;
- Support for downstream SLR activities, including metadata analysis and LLM-based study screening;
- An empirical evaluation of metadata completeness, venue characteristics, and prompting strategies for LLMs in realistic SLR scenarios.

2 Background and Related Work

The Snowballing techniques including both Forward and Backward Snowballing have been widely adopted in Software Engineering Research, due to their effectiveness in helping researchers identify relevant studies, discover emerging themes and uncover gaps in the literature. However, systematic literature review is also a highly intensive process in terms of manual and cognitive effort, often requiring a considerable amount of time to analyze citations, verify relevance, and maintain methodological consistency. Furthermore, it is also prone to biases and inconsistencies, especially when high methodological rigor and reproducibility are crucial for research production. Therefore, to address these limitations, several automated technologies have been proposed and evaluated in recent studies. Michael [5]. investigated the citation coverage provided by a broad set of citation tracking platforms, including traditional academic databases, such as Google Scholar, Scopus and Web of Science, and more recent scholarly search systems like Lens, Scite, Dimensions and Open Alex. Their study demonstrates that significant disparities exist among platforms regarding citation retrieval capabilities, particularly for forward citation searches, and identified Google Scholar, Semantic Scholar, ResearchGate, and Lens as leading options for forward citation searches, as the last one offers advanced data export resources. Additionally, Martin et al. [11] shows that Google Scholar provides the highest data coverage evaluated among 3 million citations, followed by Microsoft Academic and Dimensions, although coverage varies considerably between subject areas. Torres et al. [15] investigate bibliometric APIs in terms of functionality, metadata organization and interoperability, and propose a taxonomy for comparing search features and support for diverse bibliometric analyses. Collectively, these studies offer valuable insights into the structural and technical distinctions among bibliographic platforms and data sources, along with revealing a deficiency in the support systems for both Forward and Backward Snowballing in SLR. Nevertheless, their analyses remain focused on large-scale metrics, database coverage and system characteristics, without examining how such differences affect practical SLR activities. Consequently, there is still limited empirical evidence regarding the influence of tool-specific variations on the actual SLR settings.

3 SnowMap tool

The tool was implemented as a Python-based application and integrates multiple scholarly data sources through public APIs, including *OpenAlex*, *Semantic Scholar*, and *Crossref*. The decision to rely on these APIs was directly informed by the findings of our comparative analysis [8], which indicated that tools exposing structured and programmatic access tend to provide higher-quality metadata, better reproducibility, and greater flexibility for integration into automated workflows.

The tool was designed around three main functionalities: (i) *Citation Mapping*. SnowMap performs forward and backward snowballing starting from one or more seed studies provided by the researcher. It automatically retrieves citing articles and constructs a citation network that captures the relationships between publications. This representation supports researchers in identifying influential studies, recurring authors, and emerging research clusters within the domain. (ii) *Initial Analysis of Retrieved Studies*. The retrieved dataset can be exported in a structured format and is accompanied by an analytical summary of key metadata attributes. This includes descriptive statistics such as publication year distribution, venue frequency, quantity of open access articles, the top 3 cited papers, total of articles and total of venues. These analytics provide researchers with a quick overview of the retrieved corpus, facilitating the identification of trends, gaps, and potential biases in the literature. (iii) *Automated Screening Based on Inclusion and Exclusion Criteria*. Given a set of criteria defined by the researcher, SnowMap performs the screening of studies using LLM-based classification. Based on metadata information, studies are categorized as potentially relevant or irrelevant. This functionality is based on recent evidence [3] demonstrating the potential of LLMs to assist in the study selection phase. By leveraging LLMs, the tool aims to reduce the effort required in the early filtering stages while maintaining flexibility for human validation.

3.1 Architecture and Design Decisions

Figure 1 illustrates the sequence diagram of the SnowMap system, highlighting the interaction among its main components. The SnowMap Web App orchestrates the workflow by coordinating requests and integrating system modules.

Data discovery is performed through RESTful APIs (CrossRef and Semantic Scholar), enabling asynchronous retrieval and reconstruction of the citation graph from seed articles. The business logic layer uses conditional control structures (alt blocks) to manage execution paths across analysis stages.

The LLM Engine performs automated screening based on inclusion and exclusion criteria. It operates iteratively (loop), applying natural language processing techniques to classify studies as relevant or irrelevant.

SnowMap follows a three-tier architecture composed of a front-end interface, a back-end web server, and a data processing layer, designed to separate concerns and allow each component to be maintained and extended independently.

The front-end was built with HTML5, CSS3, and JavaScript, using Bootstrap as the UI framework to ensure a responsive and consistent visual design. The interface is structured across three pages, the Index Page, the Data Analysis Page, and the Inclusion and

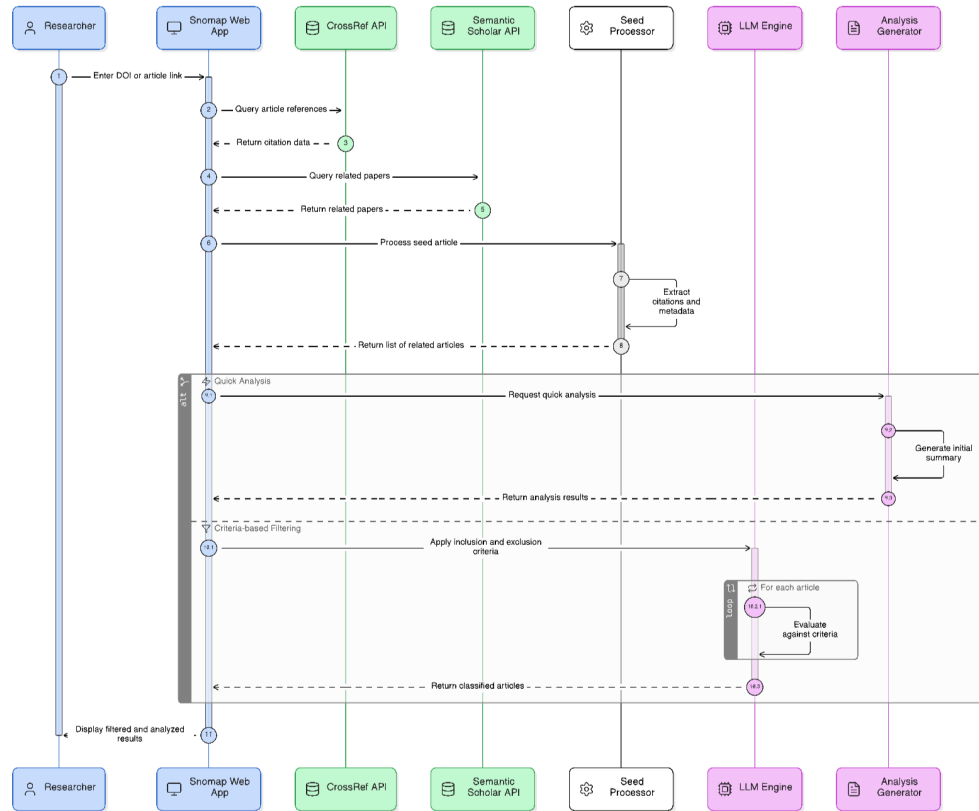


Figure 1: Sequence diagram of the SnowMap system.

Exclusion Page, each corresponding to a distinct stage of the snowballing workflow. Communication with the back-end is performed via asynchronous HTTP requests. Notably, forward and backward snowballing queries are dispatched as two independent parallel requests upon user submission, rather than sequentially, which reduces overall response time and improves the user experience during searches.

The back-end is built on Node.js with the Express framework, exposing a REST API that serves as the intermediary between the front-end and the data processing layer. Search requests are forwarded to a Python-based processing module, which is responsible for querying three external scholarly APIs, Semantic Scholar, OpenAlex, and Crossref, and combining their results. Deduplication is performed by DOI and normalized title to maximize coverage while avoiding redundant entries. A local SQLite database operates as a caching layer, storing previously retrieved results indexed by DOI and snowballing direction, so that repeated queries on the same seed study bypass external API calls entirely.

LLM-based screening is handled by a dedicated module invoked through a separate endpoint, powered locally via Ollama with automatic fallback between models in case of unavailability.

3.2 Main Functionalities and User Flow

The SnowMap system consists of three main interfaces: the Index Page (Fig. 2, item 1), the Data Analysis Page (Fig. 2, item 3), and the Inclusion and Exclusion Criteria Page (Fig. 2, item 4). The Index Page enables article retrieval using title or DOI, as well as the selection of the snowballing strategy to be applied. After query execution, the system presents SLR metadata and a structured table containing the retrieved studies associated with the seed article (Fig. 2, item 2). From this set, articles can be selected for export in CSV format or forwarded to subsequent analysis stages.

The Data Analysis Page (Fig. 2, item 3) provides a dashboard summarizing the main characteristics of the retrieved corpus, including the total number of articles, number of distinct venues, number of open-access publications, most cited papers, as well as temporal and venue-based distributions. The visualizations are interactive and dynamically updated according to user selections. In particular, filtering by publication year automatically updates the venue distribution, restricting the analysis to the selected time range. To ensure data consistency, the interface prevents the deselection of all years simultaneously.

The Inclusion and Exclusion Criteria Page (Fig. 2, item 4) allows the definition of filtering rules for the previously selected studies. SnowMap applies these criteria through an LLM-based classification

mechanism, categorizing studies as potentially relevant or irrelevant and presenting the results in a structured table for further analysis.

4 Evaluation

Case Studies Used to Build our Ground Truth Dataset. The selection of the SLR case studies followed a purposive sampling strategy to ensure both relevance and reproducibility. The inclusion criteria were defined as follows: (i) *Use of forward snowballing*: the study must explicitly report the application of forward snowballing as part of its search strategy; (ii) *Public availability of review data*: the SLR must provide access to its dataset (e.g., seed studies, included/excluded papers), enabling replication of the forward snowballing process; (iii) *Relevance and quality*: the SLR must address topics within the SE domain and be published in well-recognized venues, ensuring methodological rigor and credibility; (iv) *Availability of labeled data*: the dataset must include annotated information to distinguish relevant from non-relevant studies, allowing comparison with automatic-generated results.

Although the selected set of SLRs is not exhaustive, it was designed to provide a sample of real-world scenarios that allow the evaluation of tool behavior across different topics, publication venues, and dataset characteristics. Also, it must ensure that all analyses can be reproduced and independently verified. Table 1 identifies the four SLRs used as case studies.

Table 1: SLRs used as case studies in the evaluation.

ID	Title	Reference	Year
SLR1	Learning Software Configuration Spaces: A Systematic Literature Review	Pereira et al. [13]	2021
SLR2	Knowledge Management in Software Testing: A Systematic Snowball Literature Review	Wnuk and Garrepalli [16]	2018
SLR3	Successful Combination of Database Search and Snowballing for Identification of Primary Studies in Systematic Literature Studies	Wohlin et al. [18]	2022
SLR4	Knowledge Management Diagnostics in Software Development Organizations: A Systematic Literature Review	Maciel et al. [9]	2018

5 Results and Discussion

We discuss the results of our research questions (RQs) as follows.

5.1 RQ₁: What types of article metadata can be extracted using Snowmap in forward and backward snowballing, and how complete and consistent are these data?

The availability and quality of metadata are fundamental in forward snowballing, as they impact the traceability of studies, study selection, and the automation of the review process [2, 10, 12]. The main fields include DOI, title, year, language, keywords, references, and citations. Tools such as ResearchRabbit, Litmaps, and Google Scholar provide only partial coverage and poorly structured data, which limits their use in systematic processes [8].

The analysis of metadata completeness in SnowMap’s backward snowballing reveals important variations across the different SLRs and evaluated fields. Structural fields such as DOI and publication year generally exhibit high completeness across most scenarios,

often close to 100%, indicating good consistency in the identification of retrieved studies. In contrast, descriptive fields such as abstract and venue show greater variability, with high missing rates in some SLRs (reaching values close to 96%–100% for abstracts in certain cases). The cited by field is not consistently populated, showing variability depending on the retrieved dataset. Overall, the results indicate that, although SnowMap maintains strong robustness for structural metadata in backward snowballing, the completeness of descriptive attributes remains more sensitive to the characteristics of the underlying bibliographic data sources.

Table 2: Percentage of missing metadata fields for SLRs.

SLR	Forward					Backward				
	DOI	Abs.	Venue	Cited	Year	DOI	Abs.	Venue	Cited	Year
SLR 1	0	32	1	38	0	0	98	20	0	0
SLR 2	0	10	3	23	0	0	100	0	0	0
SLR 3	0	14	0	30	0	0	100	0	0	0
SLR 4	0	31	0	46	0	0	96	16	0	0

5.2 RQ₂: Do the studies retrieved by the tools originate from recognized scientific sources in the SE domain?

Although automated tools can significantly expand the set of studies retrieved in SLR, ensuring the scientific credibility of these results remains essential. A high volume of publications that are either low-quality or not peer-reviewed may weaken the reliability of the review outcomes and increase the workload during study selection.

To assess the quality of the retrieved literature, we examine publication venues using widely adopted bibliometric measures, such as the SCImago Journal Rank (SJR) [4], Journal Citation Reports (JCR) [4], and the Journal Impact Factor (JIF) [4]. These indicators provide indirect measures of journal and conference influence based on citation behavior and database indexing coverage. Our analysis considers venue stratification (e.g., SJR quartiles), impact-related metrics, and inclusion in recognized indexing systems. By combining these aspects, we evaluate whether the tools preferentially retrieve studies published in reputable scientific venues consistent with the quality expectations of systematic reviews.

The forward and backward snowballing processes were analyzed to assess how effectively the tools retrieve studies based on citation relationships with the seed papers.

Table 3: Percentage distribution across forward and backward snowballing for four SLRs

SLR	Forward Snowballing					Backward Snowballing				
	Q1	Q2	Q3	Q4	No match	Q1	Q2	Q3	Q4	No match
SLR 1	79.5	7.7	4.3	7.7	0.9	48.2	10.8	3.6	15.7	21.7
SLR 2	59.5	27.0	2.7	8.1	2.7	100.0	0.0	0.0	0.0	0.0
SLR 3	71.0	17.0	6.9	4.9	0.3	77.8	16.7	0.0	5.6	0.0
SLR 4	50.0	20.0	0.0	30.0	0.0	68.2	4.5	0.0	9.1	18.2

The results indicate a concentration of studies in Q1-ranked journals across all SLRs in forward snowballing, ranging from 50.0

In backward snowballing, a similar pattern is observed, but with greater dispersion. The share of Q1 varies between 48.2% and

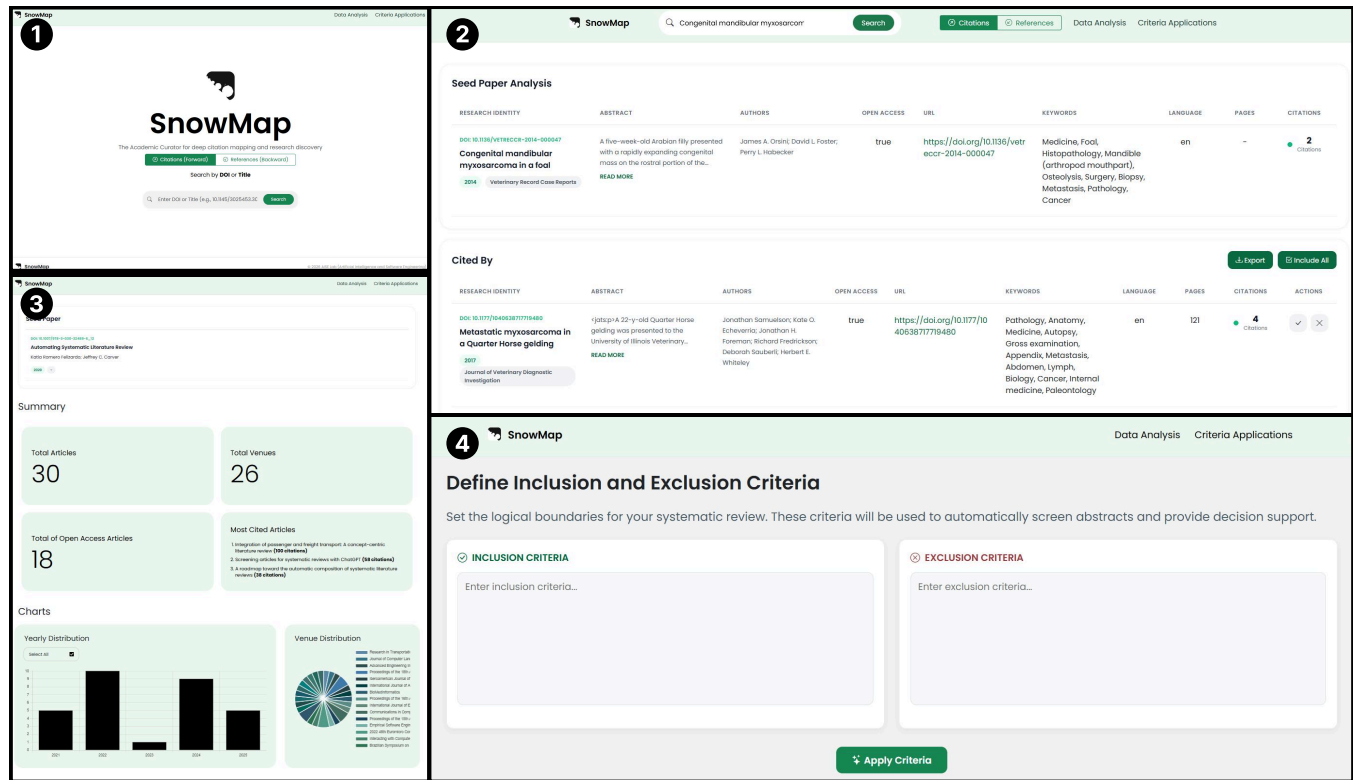


Figure 2: SnowMap’s all pages

100.0%, with a higher contribution from Q2–Q4 in some SLRs and a larger proportion of unmatched records (up to 21.7%), indicating greater sensitivity to metadata limitations and reference retrieval constraints.

Overall, SnowMap demonstrates effectiveness in retrieving high-impact studies. However, backward snowballing exhibits greater variability and lower consistency in record matching.

5.3 RQ₃: To what extent do the studies retrieved by each tool align with the ground truth in terms of coverage?

To evaluate the capability of LLMs to perform article screening based on inclusion and exclusion criteria, three prompting strategies and three distinct models were investigated in order to achieve robust performance for this task. In this section, we present and analyze the obtained results, as summarized in Table 4, and discuss the integration of this functionality into the proposed tool. Two benchmark studies were selected for evaluation [14] [1], each of which had previously undergone a manual screening process, enabling a direct comparison between the automated results and the manually obtained outcomes. The complete prompts and all experimental results, including the classifications produced for each inclusion and exclusion criterion, are publicly available in our replication package [7]. The results of the conducted experiments, as well as the scripts and datasets used in this study, are publicly available in our replication package [7].

For the LLM-based analysis, the title and abstract of each candidate article were provided as input. The models were then instructed to classify each article as either “yes” or “no” with respect to each criteria. Three prompting strategies were employed, namely zero-shot, few-shot, and chain-of-thought prompting. Among them, the chain-of-thought approach achieved the best overall performance, yielding F1-scores of 0.72 and 0.64 for [14] and [1], respectively. These findings suggest that the effective application of prompting strategies can significantly enhance the suitability of LLMs for supporting article screening tasks in systematic studies.

Table 4: Performance of the evaluated LLMs and prompts in the automated screening phase

Model	Prompt	Pereira et al. [13]			Huotala et al. [6]		
		Precision	Recall	F1-score	Precision	Recall	F1-score
gemini-3.1-flash-lite	Zero-shot	0.38	0.38	0.38	0.33	0.41	0.37
	Few-shot	0.60	0.40	0.48	0.37	0.48	0.42
	CoT	0.60	0.46	0.52	0.33	0.56	0.42
gemma4:31b	Zero-shot	0.41	0.30	0.35	0.21	0.29	0.24
	Few-shot	0.47	0.35	0.40	0.24	0.33	0.28
	CoT	0.50	0.38	0.43	0.26	0.37	0.31
ChatGPT (GPT-5.5)	Zero-shot	0.64	0.55	0.59	0.42	0.58	0.49
	Few-shot	0.71	0.63	0.67	0.51	0.66	0.57
	CoT	0.76	0.69	0.72	0.58	0.71	0.64

6 Limitations

Despite the promising results, SnowMap still presents technical limitations mainly related to its dependence on external bibliographic

APIs, such as Semantic Scholar, OpenAlex, and Crossref. Availability issues, rate limits, and inconsistencies across sources may affect record retrieval and the reconstruction of citation relationships. In addition, the study matching process relies heavily on identifiers such as DOI, meaning that incomplete or inconsistent records may lead to duplicates, missing citations, or failures in the snowballing chain, particularly in backward snowballing.

The tool also presents scalability limitations in very large-scale systematic reviews due to the high number of API requests and the processing cost associated with citation graph construction and analysis. Furthermore, the LLM-based screening module is sensitive to both the selected model and the prompting strategy, potentially producing inconsistent responses for studies with ambiguous or insufficiently detailed abstracts, thus still requiring human validation. Nevertheless, the results demonstrate that SnowMap is capable of effectively supporting automated snowballing workflows in realistic SLR scenarios.

7 Conclusion

This study presented SnowMap, a tool designed to support automated forward and backward snowballing in SLRs, as well as downstream tasks such as metadata analysis and LLM-based study screening. The results indicate that the tool is able to support an integrated pipeline for literature retrieval and processing, achieving high consistency in the extraction of structural metadata, such as DOI and publication year, across both snowballing directions. However, more descriptive metadata, including abstracts, venue information, and citation counts, exhibits greater variability, particularly in backward snowballing, reflecting the reliance on heterogeneous and, in some cases, incomplete external bibliographic data sources. In addition, the retrieved studies tend to be concentrated in high-impact venues, although backward snowballing shows greater dispersion across publication quality and a higher proportion of unmatched records.

Regarding LLM-based screening, the results show that model performance is sensitive to the prompting strategy, with chain-of-thought approaches achieving better results, suggesting that structured reasoning improves decision-making in inclusion and exclusion tasks.

As future work, we plan to enhance the robustness of SnowMap by integrating additional bibliographic data sources and metadata enrichment mechanisms, as well as investigating more advanced retrieval and record disambiguation strategies. We also aim to explore more recent language models and prompting techniques to improve screening accuracy, and to evaluate the system's effectiveness in real-world large-scale SLR scenarios.

Artifact Availability and License

The source code for SnowMap is available at [7] under the license MIT License.

References

- [1] Silvio Alonso, Marcos Kalinowski, Marx Viana, Bruna Ferreira, and Simone D.J. Barbosa. 2021. A Systematic Mapping Study on the Use of Software Engineering Practices to Develop MVPs. In *2021 47th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*. 62–69. doi:10.1109/SEAA53835.2021.00017
- [2] Katia R Felizardo and Jeffrey C Carver. 2020. Automating systematic literature review. *Contemporary empirical methods in software engineering* (2020), 327–355. doi:10.1007/978-3-030-32489-6_12
- [3] Katia Romero Felizardo, Márcia Sampaio Lima, Anderson Deizepe, Tayana Uchôa Conte, and Igor Steinmacher. 2024. ChatGPT application in systematic literature reviews in software Engineering: an evaluation of its accuracy to support the selection activity. In *Proceedings of the ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)* (Barcelona, Spain) (ESEM '24). Association for Computing Machinery, New York, NY, USA, 25–36. doi:10.1145/3674805.3686666
- [4] Borja Gonzalez-Pereira, Vicente P. Guerrero-Bote, and Félix Moya-Anegón. 2010. A new approach to the metric of journals' scientific prestige: The SJR indicator. *Journal of Informetrics* 4, 3 (2010), 379–391. doi:10.1016/j.joi.2010.03.002
- [5] Michael Gusenbauer. 2024. Beyond google scholar, scopus, and web of science: an evaluation of the backward and forward citation coverage of 59 databases' citation indices. *Research Synthesis Methods* 15, 5 (2024), 802–817.
- [6] Aleks Huotala, Miikka Kuuttila, and Mika M"antyl"a. 2025. SESR-Eval: Dataset for Evaluating LLMs in the Title-Abstract Screening of Systematic Reviews. In *2025 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*. 01–12. doi:10.1109/ESEM64174.2025.00053
- [7] Jailma Januário, Alice Fico, Julia Simão, Laiza Agatha, Fernanda Piquet, Ricardo Ramalho, and Juliana Alves Pereira. 2026. Replication Package. https://github.com/aisepucro/forward-snowballing/tree/a4220bd7d52f80e92f2f5762dac260a758174ce9/Framework_snowballing/.
- [8] Jailma Januário, Maria Nicolau, Katia Felizardo, and Juliana Pereira. 2025. Toward Reliable Forward Snowballing in Systematic Literature Reviews: A Comparative Study and Framework Proposal. In *Anais do XXXIX Simpósio Brasileiro de Engenharia de Software* (Recife/PE). SBC, Porto Alegre, RS, Brasil, 762–768. doi:10.5753/sbes.2025.11559
- [9] Claudia PC Maciel, Érica Ferreira de Souza, Ricardo de Almeida Falbo, Katia Romero Felizardo, and Nandamudi L Vijaykumar. 2018. Knowledge management diagnostics in software development organizations: a systematic literature review. In *Proceedings of the XVII Brazilian Symposium on Software Quality*. 141–150.
- [10] Iain J. Marshall and Byron C. Wallace. 2019. Toward systematic review automation: a practical guide to using machine learning tools in research synthesis. *Systematic Reviews* 8, 163 (2019). doi:10.1186/s13643-019-1074-9
- [11] Alberto Martín-Martín, Mike Thelwall, Enrique Orduna-Malea, and Emilio Delgado López-Cózar. 2021. Google Scholar, Microsoft Academic, Scopus, Dimensions, Web of Science, and OpenCitations' COCI: a multidisciplinary comparison of coverage via citations. *Scientometrics* 126, 1 (2021), 871–906.
- [12] Bianca Minetto Napoleão, Fabio Petrillo, and Sylvain Hallé. 2021. Automated support for searching and selecting evidence in software engineering: A cross-domain systematic mapping. In *2021 47th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*. IEEE, 45–53.
- [13] Juliana Alves Pereira, Mathieu Acher, Hugo Martin, Jean-Marc Jézéquel, Goetz Botterweck, and Anthony Ventresque. 2021. Learning software configuration spaces: A systematic literature review. *Journal of Systems and Software* 182 (2021), 111044.
- [14] Szymon Stradowski and Lech Madeyski. 2023. Industrial applications of software defect prediction using machine learning: A business-driven systematic literature review. *Information and Software Technology* 159 (2023), 107192. doi:10.1016/j.infsof.2023.107192
- [15] Daniel Torres-Salinas, Nicolas Robinson-Garcia, and Enrique Herrera-Viedma. 2023. New trends in bibliometric APIs: a comparative analysis. *Journal of Informetrics* 17, 3 (2023), 103385. doi:10.1016/j.ipm.2023.103385
- [16] Krzysztof Wnuk and Thrinay Garrepalli. 2018. Knowledge management in software testing: a systematic snowball literature review. *e-Informatica Software Engineering Journal* 12, 1 (2018).
- [17] Claes Wohlin. 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *EASE '14: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering*. ACM, 1–10. doi:10.1145/2601248.2601268
- [18] Claes Wohlin, Marcos Kalinowski, Katia Romero Felizardo, and Emilia Mendes. 2022. Successful combination of database search and snowballing for identification of primary studies in systematic literature studies. *Information and Software Technology* 147 (2022), 106908. doi:10.1016/j.infsof.2022.106908